

PROJECT DATA SHEET

**CASE Consultants**

Civil And Structural Engineers

Moose Hall

Barewell Road

Torquay

TQ1 4PA

tel. (01803) 200175

fax. (01803) 323205

E-Mail: admin@caseconsultants.net

JOB/PROJECT : ALTERATIONS TO MANOR CORNER, PAIGNTON**JOB/PROJECT NO :** C8030 **SHEET NO :** COVER**GENERAL :****CLIENT :** MR P.WOODWARD**ARCHITECT :** EJW ARCHITECTS**AUTHORITY :** TORBAY COUNCIL**JOB/PROJECT DESCRIPTION :** INSTALLATION OF NEW FLAT ROOF**PROPOSED USE:** COMMERCIAL**OVERALL SIZE :** AS CALCS**No. OF STOREYS :** 2**TYPE OF SOIL INVESTIGATION :** N/A**BEARING STRATA :** EXISTING MASONRY PERIMETER WALL**TYPE OF FOUNDATION :** N/A**MAXIMUM SAFE BEARING PRESSURE** N/A**STRUCTURAL MATERIALS USED :** TIMBER GRADE C24, S275 STEEL, JJI JOISTS**STANDARDS AND CODES USED IN DESIGN :**

General actions: Densities, self weight & imposed loads	BS EN 1991-1-1	☒	Steelwork	BS EN 1993 (EC3)	☒
			Timber	BS EN 1995 (EC5)	☒
General actions: Actions on structures exposed to fire	BS EN 1991-1-2	☐	Masonry	BS EN 1996 (EC6)	☐
Snow Loads	BS EN 1991-1-3	☐	Other :		☐
Wind Loads	BS EN 1991-1-4	☐	Other :		☐
Concrete	BS EN 1992 (EC2)	☐	Other :		☐

AUTHORISATION OF CALCULATION SHEETS AND BENDING SCHEDULES :

CALCULATION SHEET NOS.	CHECKED BY	DATE
Sheet nos: R01 - R22 inclusive	<i>A.C. HEARD</i>	<i>9/05/24</i>



Calculations

Contract	ALTERATIONS TO MAJOR CORDEN PAIGNTON	Job No.	C8030	Sheet No.	R01
		Designed by	D.P.O.H	Date:	MAY 2024.

DESCRIPTION

These calculations cover the design of the structural members required to undertake construction of a flat roof. The central wall is non load bearing and therefore can't be used to support roof joists without strengthening.

These calculations give three design options, option 1 uses a timber truss to clear span from elevation walls, Option 2 uses steel beams spanning from wall to wall with timber beams to split the span of the joists, Option 3 uses JJI Joists spanning to the roof midpoint with central beams on posts to pick them up.

All information has been taken from ejw Architects drawings and should be verified by further investigation prior to fabrication. All dimensions are for design purpose only.

CONTENTS:

PAGE No:

LOADING	02
OPTION 1	03 - 11
OPTION 2	12 - 14
OPTION 3	15 - 18
PLANS	19 - 20
SECTIONS	21 - 22



Calculations

Contract	ALTERATIONS TO MAJOR CORNER PAIGNTON	Job No.	C8030	Sheet No.	R02
		Designed by	D.P.H	Date:	MAY 2024

LOADING:

FLAT ROOF:

$$\begin{aligned} \text{Single Ply} &= 0.05 \text{ kN/m}^2 \\ \text{18mm Deck} &= 0.15 \text{ kN/m}^2 \\ \text{Structural Selfweight} &= 0.20 \text{ kN/m}^2 \end{aligned}$$

$$\Sigma = 0.40 \text{ kN/m}^2 \quad \times \quad 1.35 = 0.54 \text{ kN/m}^2$$

$$LL = 0.60 \text{ kN/m}^2 \quad \times \quad 1.50 = 0.90 \text{ kN/m}^2$$

$$\Sigma = 1.00 \text{ kN/m}^2 \quad \Sigma = 1.44 \text{ kN/m}^2$$

For option 1 - Truss design taking trusses @ 400 CRS

$$UDL = \text{Dead Load} = 0.54 \times 0.4 = 0.22 \text{ kN/m}$$

$$\text{Live Load} = 0.90 \times 0.4 = 0.36 \text{ kN/m}$$



Calculations

Contract

ALTERATIONS TO MAJOR CORNER
PATERNON

Job No.

C8030

Sheet No. R03

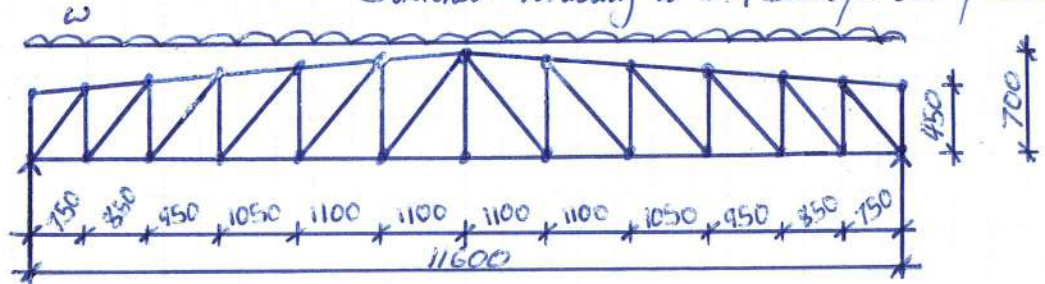
Designed by

D.P.H

Date: APRIL 2024

TRUSS DESIGN - OPTION 1

Stretched vertically at 2:1 scale for ease of understanding



Following Qualframe Analysis Pages 04-11 for live loads only:

Max compression in strut = 10.6 kN

Max tension in tie = 10.6 kN

Max shear force in strut = 0.2 kN

Max moment in strut = 0.05 kNm

Max truss deflection = 39.4 mm

∴ USE 38 x 97 C24 TIMBER FOR TRUSS WEB
AND 38 x 147 C24 TIMBER FOR TRUSS TOP
AND BOTTOM CHORDS TO SPAN BETWEEN ELEVATION
WALLS @ 400 CRS FOR OPTION 1

CASE CONSULTANTS

MOOSE HALL
BAREWELL ROAD
TORQUAY
TQ1 4PA

Printed : 11/04/2024

Sheet : *204*

File : Truss-700 Height

Title : Alterations to Manor Corner
Paignton
C8030

Designed : D.P.H

Checked :

Section groups

Truss - Timber - 38 x 97 C24



Depth = 97.0 mm
E = 2.000 kN/mm²
I = 289.0 cm⁴
A = 36.86 cm²

Truss-Chord - Timber - 38 x 147 C24

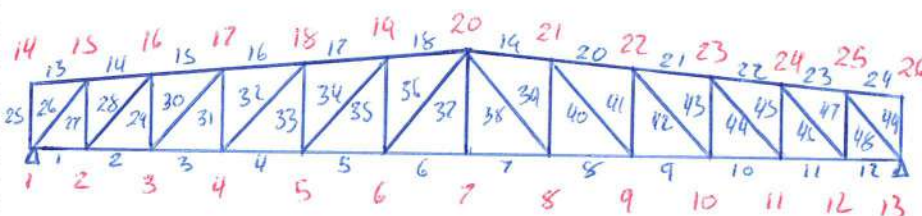


Depth = 147.0 mm
E = 2.000 kN/mm²
I = 1006 cm⁴
A = 55.86 cm²

Joints

Name	X m	Y m
J1	0.000	0.000
J2	0.750	0.000
J3	1.600	0.000
J4	2.550	0.000
J5	3.600	0.000
J6	4.700	0.000
J7	5.800	0.000
J8	6.900	0.000
J9	8.000	0.000
J10	9.050	0.000
J11	10.000	0.000
J12	10.850	0.000
J13	11.600	0.000
J14	0.000	0.350
J15	0.750	0.382
J16	1.600	0.419
J17	2.550	0.460
J18	3.600	0.505
J19	4.700	0.553
J20	5.800	0.600
J21	6.900	0.553
J22	8.000	0.505
J23	9.050	0.460
J24	10.000	0.419
J25	10.850	0.382
J26	11.600	0.350

*Joint numbers in Red
Member numbers in Blue*



Stretched vertically @ 2:1 scale for ease of understanding.

Members

Name	Length m	Section group	Ends Left/low'	Other	Joints
M1	0.750	Truss-Chord	Pin	Pin	J1 to 2
M2	0.850	Truss-Chord	Pin	Pin	J2 to 3
M3	0.950	Truss-Chord	Pin	Pin	J3 to 4
M4	1.050	Truss-Chord	Pin	Pin	J4 to 5
M5	1.100	Truss-Chord	Pin	Pin	J5 to 6

CASE CONSULTANTS

MOOSE HALL
BAREWELL ROAD
TORQUAY
TQ1 4PA

Printed : 11/04/2024

Sheet : *R05*

File : Truss-700 Height

Title : Alterations to Manor Corner
Paignton
C8030

Designed : D.P.H

Checked :

Members

Name	Length m	Section group	Ends Left/low'	Other	Joints
M6	1.100	Truss-Chord	Pin	Pin	J6 to 7
M7	1.100	Truss-Chord	Pin	Pin	J7 to 8
M8	1.100	Truss-Chord	Pin	Pin	J8 to 9
M9	1.050	Truss-Chord	Pin	Pin	J9 to 10
M10	0.950	Truss-Chord	Pin	Pin	J10 to 11
M11	0.850	Truss-Chord	Pin	Pin	J11 to 12
M12	0.750	Truss-Chord	Pin	Pin	J12 to 13
M13	0.751	Truss-Chord	Pin	Pin	J14 to 15
M14	0.851	Truss-Chord	Pin	Pin	J15 to 16
M15	0.951	Truss-Chord	Pin	Pin	J16 to 17
M16	1.051	Truss-Chord	Pin	Pin	J17 to 18
M17	1.101	Truss-Chord	Pin	Pin	J18 to 19
M18	1.101	Truss-Chord	Pin	Pin	J19 to 20
M19	1.101	Truss-Chord	Pin	Pin	J20 to 21
M20	1.101	Truss-Chord	Pin	Pin	J21 to 22
M21	1.051	Truss-Chord	Pin	Pin	J22 to 23
M22	0.951	Truss-Chord	Pin	Pin	J23 to 24
M23	0.851	Truss-Chord	Pin	Pin	J24 to 25
M24	0.751	Truss-Chord	Pin	Pin	J25 to 26
M25	0.350	Truss	Pin	Pin	J1 to 14
M26	0.842	Truss	Pin	Pin	J1 to 15
M27	0.382	Truss	Pin	Pin	J2 to 15
M28	0.948	Truss	Pin	Pin	J2 to 16
M29	0.419	Truss	Pin	Pin	J3 to 16
M30	1.056	Truss	Pin	Pin	J3 to 17
M31	0.460	Truss	Pin	Pin	J4 to 17
M32	1.165	Truss	Pin	Pin	J4 to 18
M33	0.505	Truss	Pin	Pin	J5 to 18
M34	1.231	Truss	Pin	Pin	J5 to 19
M35	0.553	Truss	Pin	Pin	J6 to 19
M36	1.253	Truss	Pin	Pin	J6 to 20
M37	0.600	Truss	Pin	Pin	J7 to 20
M38	1.253	Truss	Pin	Pin	J20 to 8
M39	0.553	Truss	Pin	Pin	J8 to 21
M40	1.231	Truss	Pin	Pin	J21 to 9
M41	0.505	Truss	Pin	Pin	J9 to 22
M42	1.165	Truss	Pin	Pin	J22 to 10
M43	0.460	Truss	Pin	Pin	J10 to 23
M44	1.056	Truss	Pin	Pin	J23 to 11
M45	0.419	Truss	Pin	Pin	J11 to 24
M46	0.948	Truss	Pin	Pin	J24 to 12
M47	0.382	Truss	Pin	Pin	J12 to 25
M48	0.842	Truss	Pin	Pin	J25 to 13
M49	0.350	Truss	Pin	Pin	J13 to 26

Bases

Name	Joint	Horizontal kN/m	Vertical kN/m	Rotational kN.m/Rad
B1	J1	Fixed	Fixed	Pinned
B2	J13	Free	Fixed	Pinned

CASE CONSULTANTS

MOOSE HALL
BAREWELL ROAD
TORQUAY
TQ1 4PA

Printed : 11/04/2024

Sheet : *R06*

File : Truss-700 Height

Title : Alterations to Manor Corner
Paignton
C8030

Designed : D.P.H

Checked :

Load cases

Name	Type
Dead (ULT)	Other
Live (ULT)	Other

Load combination set: 'Other'

Name	Limit	Factor	Load case
Live (CHAR)	SLS	0.00	Self Weight
		0.00	Dead (ULT)
		0.67	Live (ULT)
Live (ULT)	ULS+SLS	0.00	Self Weight
		0.00	Dead (ULT)
		1.00	Live (ULT)

Uniformly distributed loads**Load case : Dead (ULT)**

Name	Member	Value kN/m	Direction
L50	M13	0.22	Vertical ↓
L51	M14	0.22	Vertical ↓
L52	M15	0.22	Vertical ↓
L53	M16	0.22	Vertical ↓
L54	M17	0.22	Vertical ↓
L55	M18	0.22	Vertical ↓
L56	M19	0.22	Vertical ↓
L57	M20	0.22	Vertical ↓
L58	M21	0.22	Vertical ↓
L59	M22	0.22	Vertical ↓
L60	M23	0.22	Vertical ↓
L61	M24	0.22	Vertical ↓

Uniformly distributed loads**Load case : Live (ULT)**

Name	Member	Value kN/m	Direction
L62	M13	0.36	Vertical ↓
L63	M14	0.36	Vertical ↓
L64	M15	0.36	Vertical ↓
L65	M16	0.36	Vertical ↓
L66	M17	0.36	Vertical ↓
L67	M18	0.36	Vertical ↓
L68	M19	0.36	Vertical ↓
L69	M20	0.36	Vertical ↓
L70	M21	0.36	Vertical ↓
L71	M22	0.36	Vertical ↓
L72	M23	0.36	Vertical ↓
L73	M24	0.36	Vertical ↓

CASE CONSULTANTS

MOOSE HALL
BAREWELL ROAD
TORQUAY
TQ1 4PA

Printed : 11/04/2024

Sheet : R07

File : Truss-700 Height

Title : Alterations to Manor Corner
Paignton
C8030

Designed : D.P.H

Checked :

Reactions - Linear elastic analysis

Loading combination: 'Live (ULT)' of combination set: 'Other'

Base	Joint	Horizontal kN	Vertical kN	Moment kN.m
B1	J1	0.00	2.09 ↑	0.00
B2	J13	-0.00	2.09 ↑	-0.00

Joint deflections - Linear elastic analysis

Loading combination: 'Live (CHAR)' of combination set: 'Other'

Joint	Dx mm	Dy mm	Pz Rad
J1	0.0	0.0	-0.0122 ↻
J2	0.2 →	-9.2 ↓	-0.0122 ↻
J3	0.5 →	-18.6 ↓	-0.0111 ↻
J4	1.0 →	-27.4 ↓	-0.0092 ↻
J5	1.7 →	-34.3 ↓	-0.0066 ↻
J6	2.4 →	-38.3 ↓	-0.0037 ↻
J7	3.0 →	-39.4 ↓	-0.0010 ↻
J8	3.7 →	-38.3 ↓	0.0010 ↻
J9	4.4 →	-34.3 ↓	0.0037 ↻
J10	5.1 →	-27.4 ↓	0.0066 ↻
J11	5.6 →	-18.6 ↓	0.0092 ↻
J12	5.9 →	-9.2 ↓	0.0111 ↻
J13	6.1 →	0.0	0.0122 ↻
J14	3.9 →	-0.0	-0.0124 ↻
J15	4.3 →	-9.1 ↓	-0.0119 ↻
J16	4.5 →	-18.6 ↓	-0.0108 ↻
J17	4.5 →	-27.3 ↓	-0.0087 ↻
J18	4.2 →	-34.3 ↓	-0.0060 ↻
J19	3.7 →	-38.4 ↓	-0.0030 ↻
J20	3.0 →	-39.4 ↓	-0.0003 ↻
J21	2.4 →	-38.4 ↓	0.0016 ↻
J22	1.9 →	-34.3 ↓	0.0044 ↻
J23	1.6 →	-27.3 ↓	0.0072 ↻
J24	1.6 →	-18.6 ↓	0.0096 ↻
J25	1.8 →	-9.1 ↓	0.0114 ↻
J26	2.2 →	-0.0	0.0124 ↻

MAX TROSS DEFLECTION
(LIVE LOAD ONLY)

Forces - Linear elastic analysis

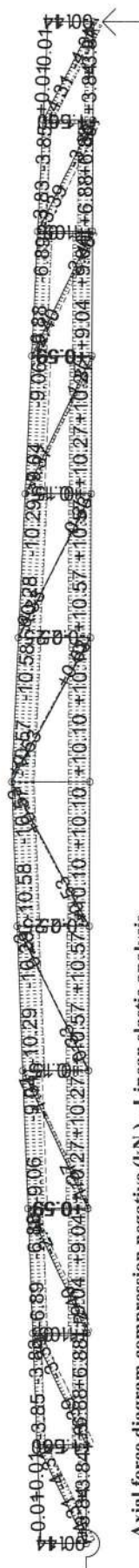
Loading combination: 'Live (ULT)' of combination set: 'Other'

Member	Distance m	Measured from	Axial kN	Sense	Shear kN	Moment kN.m
M17	0.000	Left end	-10.29	Compression	0.20	0.00
	0.551	Left end	-10.28	Compression	-0.00	0.05
	1.101	Left end	-10.28	Compression	-0.20	0.00
M18	0.000	Left end	-10.58	Compression	0.20	0.00
	0.551	Left end	-10.58	Compression	-0.00	0.05
	1.101	Left end	-10.57	Compression	-0.20	0.00

MAX COMPRESSION
IN STRUT (LLD)

MAX SHEAR IN
STRUT (LLD)

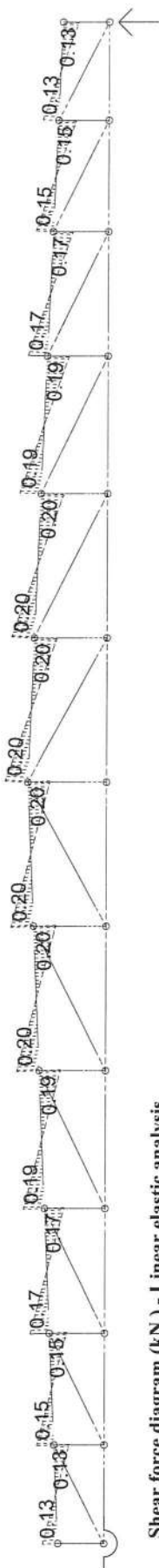
MAX MOMENT
IN STRUT
(LLD)



Axial force diagram compression negative (kN.) - Linear elastic analysis

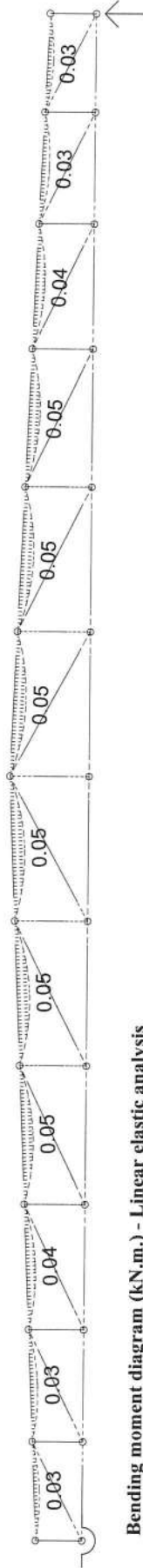
Loading combination: 'Live (ULT)' of combination set: 'Other'

CASE CONSULTANTS MOOSE HALL BAREWELL ROAD TORQUAY TQ1 4PA	Printed : 11/04/2024		Sheet : <i>RO8</i>
	File : Truss-700 Height		
	Title : Alterations to Manor Corner Paignton C8030		
	Designed : D.P.H		Checked :
<i>Quikframe - Plane Frame Analysis 9.40</i>			



Shear force diagram (kN.) - Linear elastic analysis
Loading combination: 'Live (ULT)' of combination set: 'Other'

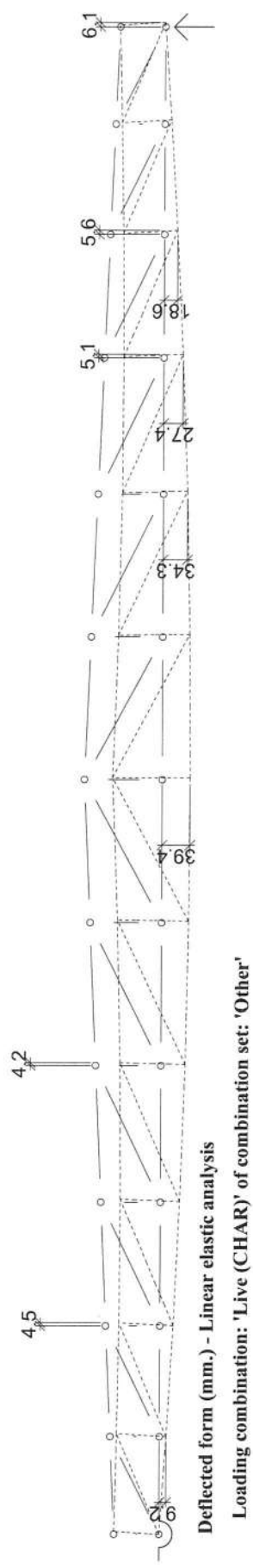
Quikframe - Plane Frame Analysis 9.40	CASE CONSULTANTS		Printed : 11/04/2024	Sheet : <i>ROA</i>
	MOOSE HALL BAREWELL ROAD TORQUAY TQ1 4PA		File : Truss-700 Height	
			Title : Alterations to Manor Corner Paignton C8030	
			Designed : D.P.H	Checked :



Bending moment diagram (kN.m.) - Linear elastic analysis
Loading combination: 'Live (ULT)' of combination set: 'Other'

CASE CONSULTANTS MOOSE HALL BAREWELL ROAD TORQUAY TQ1 4PA	Printed : 11/04/2024	Sheet : R101
	File : Truss-700 Height	
	Title : Alterations to Manor Corner Paington C8030	
	Designed : D.P.H	Checked :

Quikframe - Plane Frame Analysis 9.40



Quikframe - Plane Frame Analysis 9.40	CASE CONSULTANTS		Printed : 11/04/2024	Sheet : R11
	MOOSE HALL BAREWELL ROAD TORQUAY TQ1 4PA		File : Truss-700 Height	
			Title : Alterations to Manor Corner Paignton C8030	
			Designed : D.P.H	Checked :



Calculations

Contract	ALTERATIONS TO MAJOR CORNER PAIGUFON	Job No.	C8030	Sheet No.	R12
		Designed by	D.P.M	Date:	APRIL 2024

ROOF JOISTS - OPTION 2 - For location see sheet 19

$$\text{Design Span} = 4750 \text{ mm}$$

$$\text{UDL} = 1.44 \times 0.4 = 0.58 \text{ kN/m}$$

$$M_d = 0.58 \times \frac{4.75^2}{8} = 1.64 \text{ kNm} \quad V_A = V_B = 1.38 \text{ kN}$$

$$I_b \text{ Perm} = 24 \times 0.9 \times 11 \times 1.3 = 18.28 \text{ Nmm}^2$$

$$I_{yy} \text{ Reqd} = \frac{1.64 \times 10^6}{18.28} = 89.7 \times 10^3 \text{ mm}^3$$

$$\text{with } b=47 \quad d \text{ Reqd} = \sqrt{\frac{6 \times 89.7 \times 10^3}{47}} = 107 \text{ mm}$$

$$E \text{ Perm} = 0.003 \times 4750 = 14.3 \text{ mm (Instantaneous)}$$

$$I_{yy} \text{ Reqd} = \frac{1.01 \times 5 \times \frac{0.58}{1.4} \times 4750^4}{384 \times 11000 \times 14.3} = 19.2 \times 10^6 \text{ mm}^4$$

$$\text{with } b=47 \quad d \text{ Reqd} = \sqrt[3]{\frac{12 \times 19.2 \times 10^6}{47}} = 170 \text{ mm}$$

$$\text{try } 47 \times 195 \text{ C24 Joists giving } I_{yy} = 29.0 \times 10^6 \text{ mm}^4$$

$$\therefore \text{Initial deflection} = 14.3 \times \frac{19.2}{29.0} = 9.5 \text{ mm}$$

Consider final deflection with $\mu_{\text{eff}}=0.6$ & $\Psi_2=0$ for roof

$$\text{with } D_L = 0.40 \text{ kN/m}^2, \quad L_L = 0.60 \text{ kN/m}^2, \quad \text{total} = 1.0 \text{ kN/m}^2$$

$$\therefore \text{Final deflection} = 9.5 \times \left[\frac{(1+0.6) \times \frac{0.4}{1.0}}{6.1} \right] + 9.5 \times \left[\frac{(1+0) \times \frac{0.6}{1.0}}{5.7} \right] = 11.8 \text{ mm}$$

$$= \frac{\text{SPAN}}{403} < \frac{\text{SPAN}}{230} \therefore \text{OK}$$

$\therefore$ USE 47 X 195 C24 JOISTS @ 400 CRS.



Calculations

Contract	ALTERATIONS TO MAJOR CORNER PAIGNTON	Job No.	C8030	Sheet No.	R13
		Designed by	D.P.H	Date:	APRIL 2024

TRIMMERS - OPTION 2

Clear Span = 3000mm Design Span = 3150mm

$$UDL = \text{Dead} = 0.54 \times \left(\frac{4.75 + 3.45}{2} \right) = 2.21 \text{ kN/m}$$

$$\text{Live} = 0.90 \times \left(\frac{4.75 + 3.45}{2} \right) = 3.69 \text{ kN/m}$$

$$\Sigma = 5.90 \text{ kN/m}$$

$$M_d = 5.90 \times \frac{3.15^2}{8} = 7.32 \text{ kNm} \quad V_A = V_B = 9.29 \text{ kN}$$

$$e_{\text{Perm}} = \frac{3150}{250} = 12.6 \text{ mm}$$

Try 152 x 89 UB 16 with $I_{yy} = 834 \text{ cm}^4$

From Blue Book with $l_e = 3.15 \text{ m}$ & $C_1 = 1.13$

$$M_R = 23.1 \text{ kNm} > 7.32 \text{ kNm} \therefore \text{OK}$$

$$\text{Deflection due to Dead Load} = \frac{5 \times \frac{2.21}{1.35} \times 3150^4}{384 \times 2.05 \times 10^9 \times 834} = 1.23 \text{ mm}$$

$$\text{Deflection due to Live Load} = \frac{5 \times \frac{3.69}{1.50} \times 3150^4}{384 \times 2.05 \times 10^9 \times 834} = 1.84 \text{ mm}$$

$$\therefore \text{Total deflection} = 3.07 \text{ mm} < 12.6 \text{ mm} \therefore \text{OK}$$

$\therefore$ USE 152 x 89 UB 16 FOR TRIMMER BEAMS



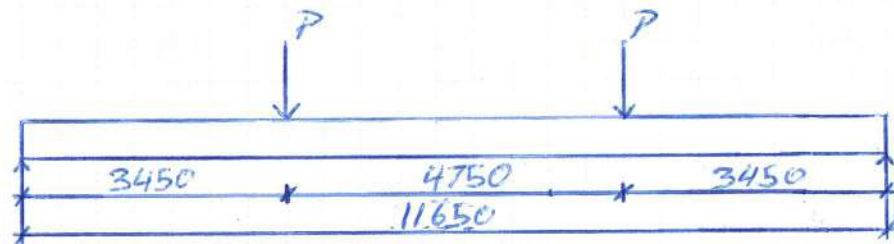
Calculations

Contract	ALTERATIONS TO HAYDOCK CORNER PAIGNTON	Job No.	C8030	Sheet No.	R14
		Designed by	D.P.H	Date:	APRIL 2024

BEAM - OPTION 2

Clear Span = 11500mm

Design Span = 11650mm



$$P = 2.21 \times \frac{3.0 + 2.25}{2} = 5.80 \text{ kN} \quad \text{Trimmer Dead Load}$$

$$3.69 \times \frac{3.0 + 2.25}{2} = 9.69 \text{ kN} \quad \text{Trimmer Live Load}$$

$$\Sigma = 15.49 \text{ kN}$$

$$M_d = 15.49 \times 3.45 = 53.4 \text{ kNm}$$

$$\epsilon_{\text{Perm}} = \frac{11650}{250} = 46.6 \text{ mm}$$

Try 305 x 127 UB 37 with $I_{yy} = 7170 \text{ cm}^4$

From Blue Book with $l_e = 4.75$ & $C_1 = 1.13$

$$M_R = 70.7 \text{ kNm} > 53.4 \text{ kNm} \quad \therefore \text{OK}$$

$$\text{Deflection due to Dead Load} = \frac{\frac{5.8}{1.35} \times 345}{24 \times 20500 \times 7170} \times (3 \times 1165^2 - (4 \times 345^2))$$

$$= 1.51 \text{ cm} = 15.1 \text{ mm}$$

$$\text{Deflection due to Live Load} = \frac{\frac{9.69}{1.5} \times 345}{24 \times 20500 \times 7170} \times (3 \times 1165^2 - (4 \times 345^2))$$

$$= 2.27 \text{ cm} = 22.7 \text{ mm}$$

$$\therefore \text{Total deflection} = 37.8 \text{ mm} < 46.6 \text{ mm} \quad \therefore \text{OK}$$

$\therefore$ USE 305 x 127 UB 37 TO CLEAR SPAN BETWEEN ELEVATION WALLS



Calculations

Contract	ALTERATIONS TO MAJOR CORNER PAIGNTON	Job No.	C8030	Sheet No.	R15
		Designed by	D.P.H	Date:	MAY 2024

RAFTERS - OPTION 3 - For location see sheet 20.

Try JJI Joists spanning to midpoint of roof.

$$\text{Design Span} = \frac{11650}{2} = 5830\text{mm}$$

$$\text{UDL} = 1.44 \times 0.4 = 0.58\text{ kN/m}$$

$$M_d = 0.58 \times \frac{5.83^2}{8} = 2.46\text{ kNm} \quad V_A = V_B = 1.69\text{ kN}$$

$$\epsilon_{\text{Perm}} = 0.003 \times 5830 = 17.5\text{mm (Instantaneous)}$$

$$EI_{\text{Reqd}} = \frac{5 \times \frac{0.58}{1.4} \times 5830^4}{384 \times 17.5} = 356.1 \times 10^9 \text{ N/mm}^2$$

Joist Type	Depth	Bending moment capacity	Bending stiffness	Shear strength capacity	Shear stiffness	Intermediate bearing capacity - minimum 89mm bearing length		End bearing capacity - minimum 45mm bearing length		End bearing capacity - minimum 89mm bearing length		Weight per metre length
		M	EI	V	GA	N/S	W/S	N/S	W/S	N/S	W/S	W
	(mm)	(kNm)	(10 ⁹ Nmm ²)	(kN)	(10 ⁶ N)	(kN)	(kN)	(kN)	(kN)	(kN)	(kN)	(kg/m)
JJI 195 A+	195	5.67	305.1	10.64	1.234	16.37	16.37	8.50	8.50	10.33	10.76	2.56
JJI 195 C		8.03	505.6	12.44	1.234	25.07	25.07	12.90	13.02	13.18	16.48	3.57
JJI 220 A+	220	6.60	407.4	11.33	1.477	16.37	16.37	8.50	8.50	10.33	10.76	2.70
JJI 220 B+		8.37	588.51	12.48	1.477	21.94	21.94	11.39	11.39	13.18	14.42	3.35

Try JJI 220 B+ giving $EI = 588.5 \times 10^9 \text{ N/mm}^2 > 356.1 \times 10^9 \text{ N/mm}^2$

$$\therefore \text{Initial deflection} = 17.5 \times \frac{356.1}{588.5} = 10.6\text{mm}$$

Consider final deflection with $k_{\text{def}} = 0.6$ & $V_2 = 0$
with $D_L = 0.46\text{ kN/m}^2$, $L_L = 0.60\text{ kN/m}^2$, Total = 1.06 kN/m^2

$$\therefore \text{Final deflection} = 10.6 \times \left[\frac{(1+0.6) \times \frac{0.47}{1.0}}{6.8} \right] + 10.6 \times \left[\frac{(1+0) \times \frac{0.6}{1.0}}{6.4} \right] = 13.2\text{mm}$$

$$= \frac{\text{SPAN}}{442} < \frac{\text{SPAN}}{250} \therefore \text{Ok} \quad \text{with } k_{\text{gys}} = 1.1 \quad k_{\text{mod}} = 0.9 \quad \& \quad \gamma_m = 1.3$$

$$\text{Moment capacity} = 8.37 \times 0.9 \times 1.1 \times \frac{1}{1.3} = 6.37\text{ kNm} > 2.46\text{ kNm} \therefore \text{Ok}$$

$$\text{Shear capacity} = 12.48 \times 0.9 \times 1.1 \times \frac{1}{1.3} = 9.50\text{ kN} > 1.69\text{ kN} \therefore \text{Ok}$$

$\therefore$ USE JJI 220 B+ JOISTS @ 400 CRS TO SPAN TO MIDPOINT OF THE ROOF



Calculations

Contract

ALTERATIONS TO MAJOR CORNER
PATEGOU

Job No.

C8030

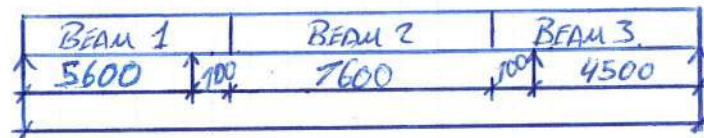
Sheet No. R16

Designed by

D.P.H

Date: May 2024

CENTRAL BEAM DESIGN - OPTION 3



Take Beams 1 & 3 as cantilevers supporting Beam 2.

Consider UDLs:

$$\text{Max UDL} = \text{Ultimate total} = 1.44 \times \frac{11.65}{2} = 8.39 \text{ kN/m}$$

$$\text{Min UDL} = \text{Characteristic dead} = 0.40 \times \frac{11.65}{2} = 2.33 \text{ kN/m}$$

Consider Moments:

taking beam connections as pinned the max moment occurs taking Beam 2 as simply supported

$$\text{Max moment} = \frac{wL^2}{8} = \frac{8.39 \times 7.6^2}{8} = 60.6 \text{ kNm}$$

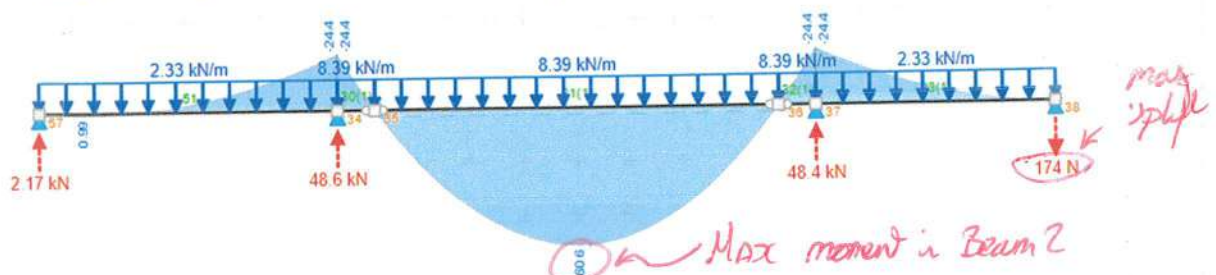
$\therefore$ try 254 x 146 UB 43 with $I_{yy} = 6540 \text{ cm}^4$ for Beam 2

from Blue Book with $l_e = 7.6$ & $C_1 = 1.13$

$$M_e = 78.0 \text{ kNm} > 60.6 \text{ kNm} \therefore \text{OK}$$

Consider Deflection:

Take worst case with max UDL between central supports and min UDL on cantilever backspan.





Calculations

Contract	ALTERATIONS TO MAJOR LORVER PITGURON	Job No.	C6030	Sheet No.	R17
		Designed by	D.P.H	Date:	MAY 2024.

$$\text{Deflection due to Beam 2} = \frac{5 \times \frac{8.39}{1.4} \times 7600^4}{384 \times 2.05 \times 10^9 \times 6540} = 19.4 \text{ mm}$$

$$\text{Deflection due to OD on cantilever and - conservatively taking no load on backspan} = \frac{\frac{8.39}{1.4} \times 700^3}{24 \times 2.05 \times 10^9 \times 4410} \times ((14 \times 5600) + (3 \times 700)) = 0.2 \text{ mm}$$

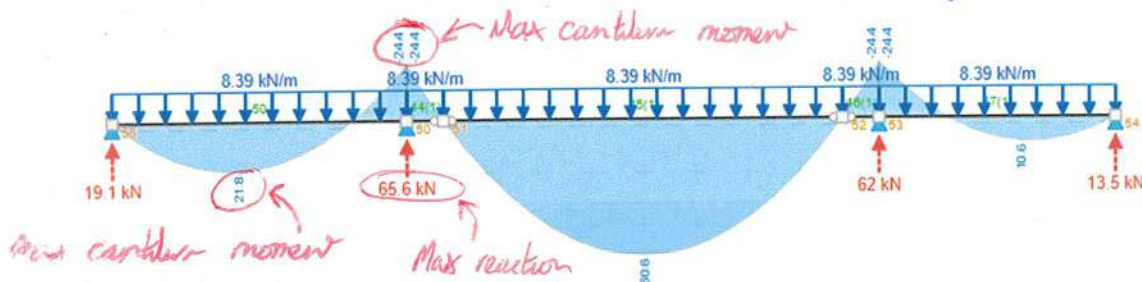
$$\text{Deflection due to point load on cantilever due to Beam 2} = \frac{\frac{31.9}{1.4} \times 700^2}{3 \times 2.05 \times 10^6 \times 4410} \times (5600 + 700) = 2.6 \text{ mm}$$

$$\therefore \text{Total deflection} = 19.4 + 0.2 + 2.6 = 22.2 \text{ mm} = \frac{\text{SPAN}}{342} < \frac{\text{SPAN}}{250} \therefore \text{OK}$$

Consider Support Reactions:

From Previous Page max tie down force required = 174 N

For Max downforce take ultimate total force everywhere.



Max downforce = 66 kN.

Consider Padstone:

$$f_b \text{ Perm} = 1.25 \times \frac{3.5}{3.0} = 1.46 \text{ N/mm}^2$$

$$\text{With } b = 215 \text{ mm} \quad L \text{ Req'd} = \frac{66 \times 10^3}{1.46 \times 215} = 210 \text{ mm}$$

Try 215mm x 440mm x 150dp GFD 3 concrete padstone

$$\text{Weight of padstone} = 22 \times 0.215 \times 0.44 \times 0.15 = 0.312 \text{ kN} > 174 \text{ N} \therefore \text{OK}$$



Calculations

Contract

ALTERATIONS TO MAJOR CORNER
PAIGNTON

Job No.

C8030

Sheet No. R18

Designed by

D.P.H

Date:

MAY 2024

∴ USE 254 x 146 UB 43 FOR BEAM 2 WITH
100 x 100 SHS 3.2 POSTS ON TO 215 wd x 440 lg
x ISOdp GEN 3 CONCRETE PADSTONES

Consider cantilever Beams:

Try 254 x 146 UB 31 with $I_{yy} = 4410 \text{ cm}^4$.

Max moment over support = 24.4 kNm

From Blue Book with $l_e = 1.5 \text{ m}$ & $C_1 = 1.13$

$M_R = 108 \text{ kNm} > 24.4 \text{ kNm} \therefore \text{OK}$
 ↑ Double length of cantilever

Max moment in Beam span = 21.8 kNm

From Blue Book with $l_e = 5.6 \text{ m}$ & $C_1 = 1.13$

$M_R = 58.6 \text{ kNm} > 21.8 \text{ kNm} \therefore \text{OK}$
 ↑ Beam clear span

Consider deflection:

Conservatively taking as simply supported beam

$$\frac{5 \times \frac{8.39}{1.4 \times 5600^4}}{384 \times 2.05 \times 10^9 \times 4410} = 8.5 \text{ mm} = \frac{\text{SPAN}}{660} < \frac{\text{SPAN}}{250} \therefore \text{OK}$$

∴ USE 254 x 146 UB 31 FOR BEAMS 1 & 3



Calculations

Contract

*ALTERATIONS TO MANOR
CORNER PAIGNTON*

Job No.

C8030

Sheet No. *R19*

Designed by

D.P.H.

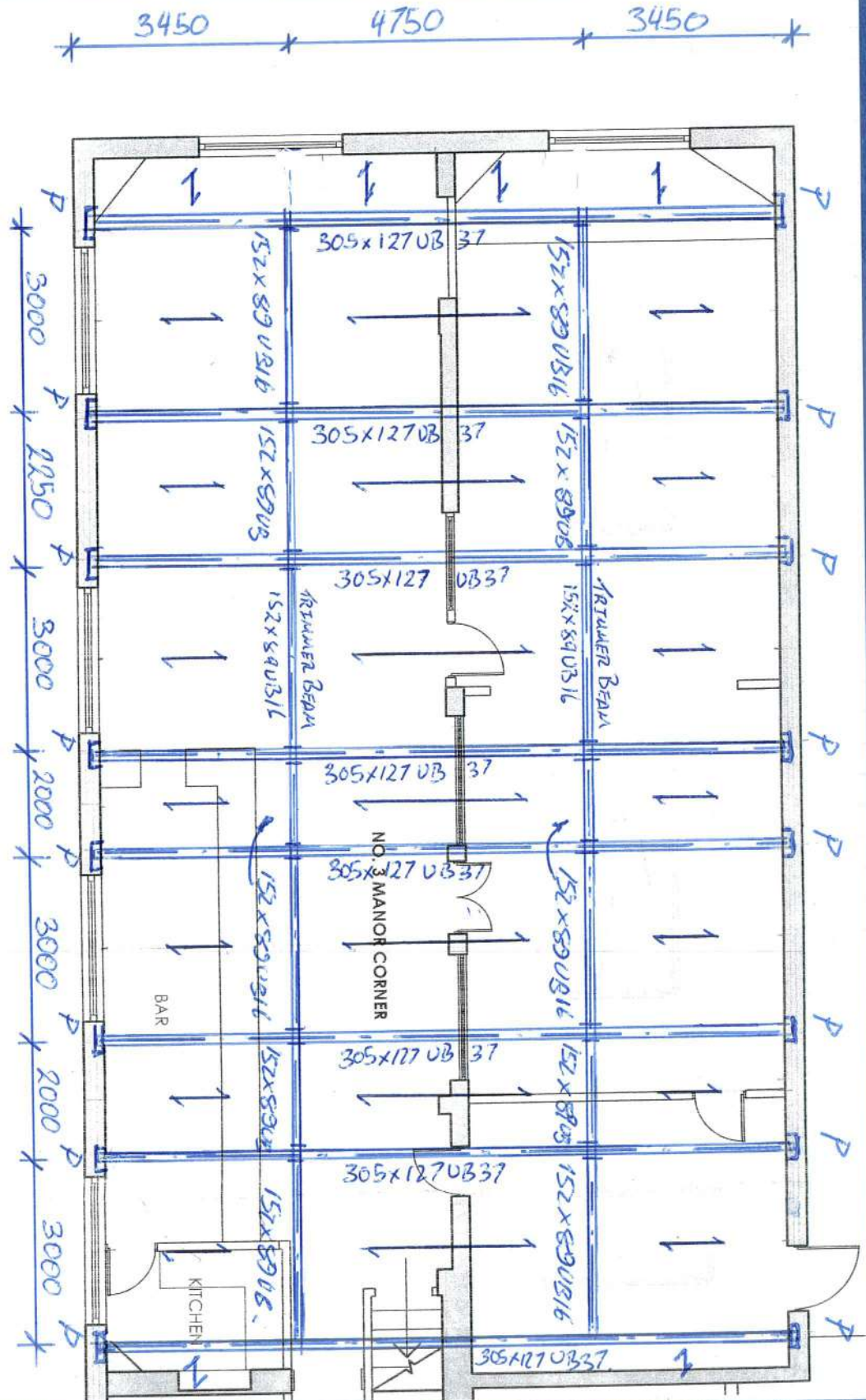
Date: *April 2024*

FIRST FLOOR 1:100

PLAN ON OPTION 2.

*INDICATES
SPAN OF
47x195
C24 JOISTS
@ 400 C/S*

*P DENOTES
325x100 u/d
x150dp GENS
PASTSTONE.*





Calculations

Contract

ALTERATIONS TO MAJOR
CORNER PORTICO

Job No.

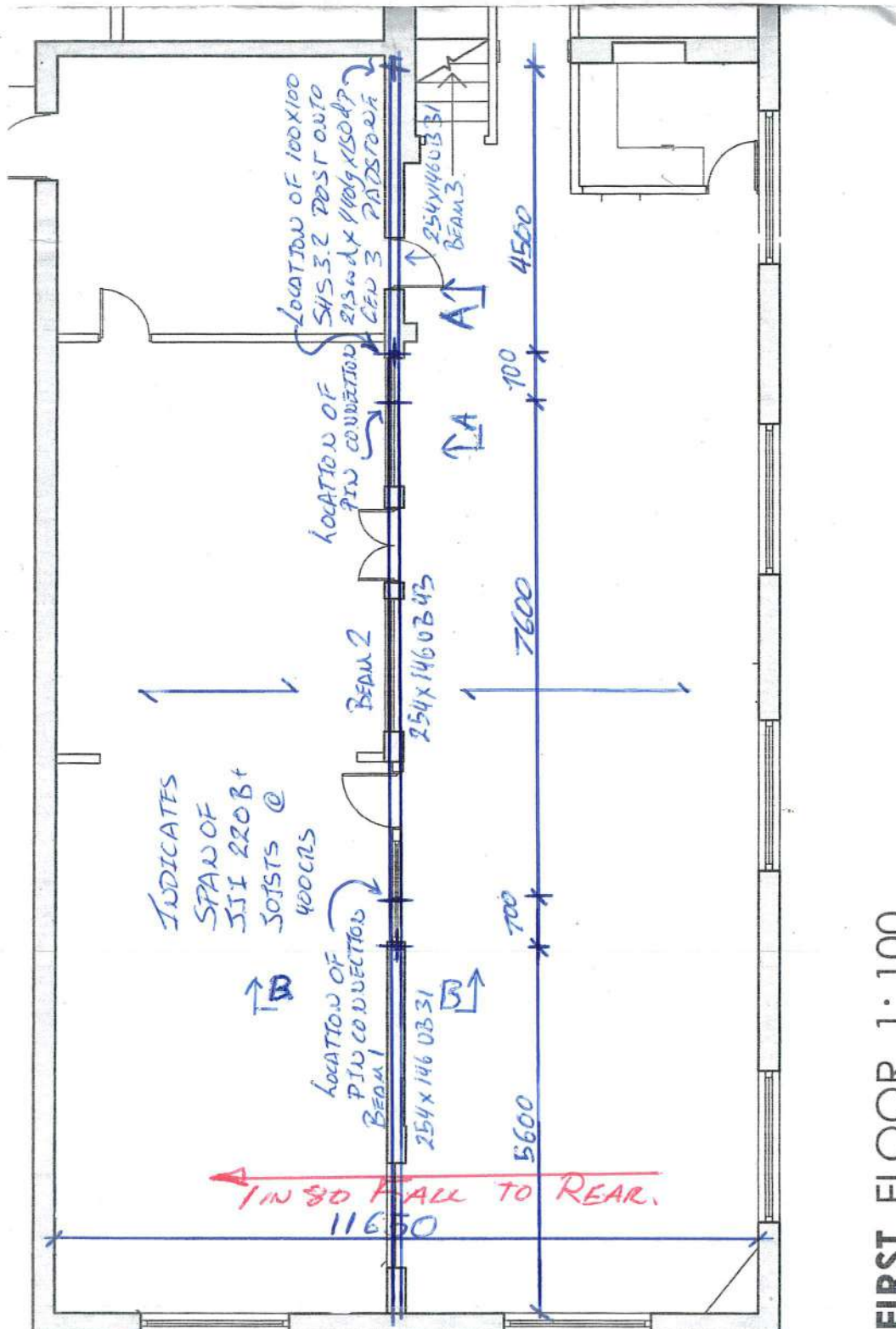
C8030

Sheet No. R20

Designed by

D.P.H

Date: May 2024.



FIRST FLOOR 1:100

Plan on Option 3



Calculations

Contract

ALTERATIONS TO HARBOUR CORNER
PAIGINTON

Job No.

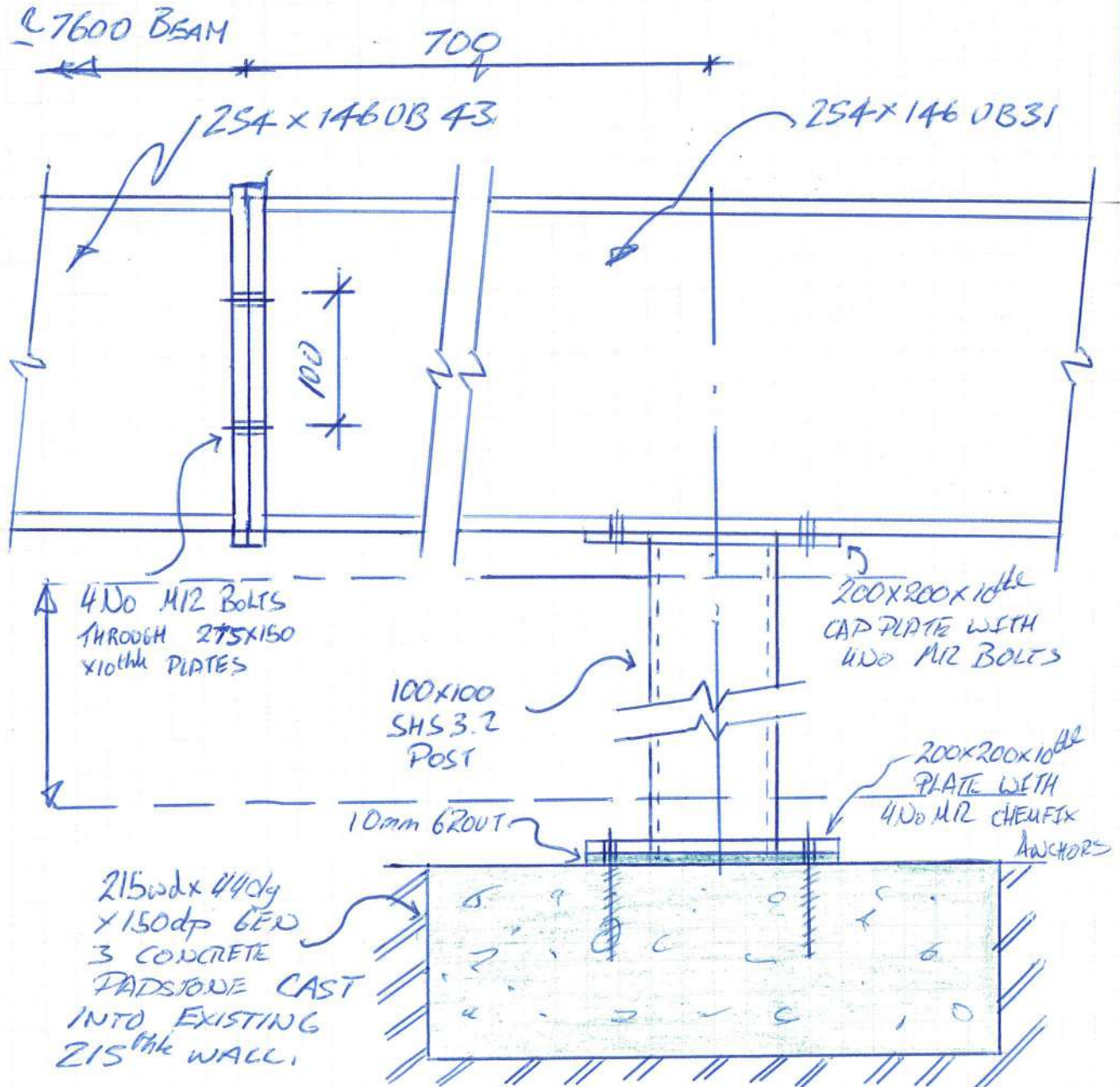
C8030

Sheet No. R21

Designed by

DOPH

Date: May 2024



SECTION A-A - OPTION 3



Calculations

Contract

ALTERATIONS TO MAJOR CARRIAGE
PAIGNTON

Job No.

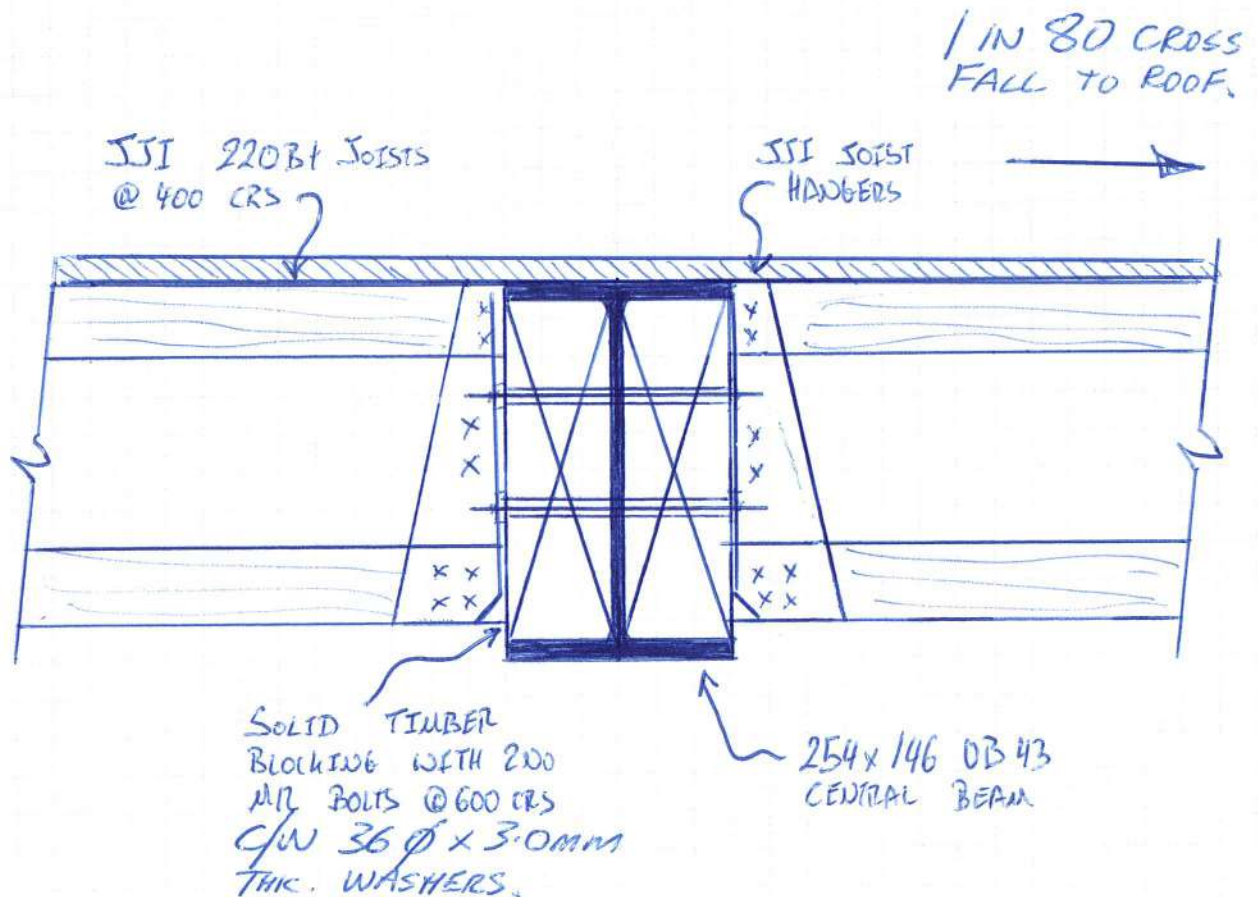
C8030

Sheet No. R22

Designed by

D.P.H.

Date: May 2024



SECTION B-B - OPTION 3